

Katharina Julia Brenner

INDUSTRIAL PROCESS MODELER + BIOENGINEER

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PROFILE

I build quantitative, plant-wide models that make complex industrial and bioengineering processes visible, comparable, and ready for better scale-up decisions.

SELECTED EXPERIENCE

2023 - NOW

PhD Candidate + Scientific Staff

Technical University of Munich - Cellular Agriculture

Industrial process modeling, bioprocess scale-up, digital twins, techno-economic and environmental decision support.

2024 - NOW

Founder

Future Foods at TUM

Building an interdisciplinary community around sustainable food, advanced technologies, research, and entrepreneurship.

2017

Class of Fall 2017

Center for Digital Technology and Management (CDTM)

Interdisciplinary innovation and technology management alongside a foundation in business administration.

SELECTED WORK

01

Cultivated Meat Process Model

Open, interactive facility model from media preparation through packaging.

02

BioTA - Bioreactor Technical Analysis

Open workflow for yield prediction, optimization, and future CFD coupling.

03

Process Intelligence Systems

Digital twins and decision interfaces connecting process physics with action.

CORE EXPERTISE

Process systems

Plant-wide simulation, mass and energy balances, facility architecture.

Scale-up

Bioreactors, utilities, bottlenecks, scenarios, TEA and LCA.

Digital engineering

Python, data science, AI, digital twins, decision tools.

Analytical foundation

Mathematics, statistics, uncertainty, optimization.

SELECTED PUBLICATIONS

2026

Decoding cultured meat manufacturing

Frontiers in Nutrition

2026

Waste stream valorization in cultured meat manufacturing

Applied Food Research

2025

Bioreactor parameters and systems for cultured meat production

Future Foods

2025

Technology and processes for cultivated meat

Ernährungs Umschau

SELECTED STATIONS

Cambridge / TUM / CDTM / Bayerische EliteAkademie / LMU / Stanford / KAIST / TUM Venture Labs

RECOGNITION

Deutschlandstipendium, 2022

Lothar-und-Sigrid-Rohde-Stiftung, 2021

Bayerische EliteAkademie, Class of 2017

Languages: German, English